1. (Fall 2024, #6) Using appropriate tests, determine whether the series below converge or diverge. Show all work, by naming the test, applying the test, and drawing the correct conclusion.

(a) $\sum_{n=1}^{\infty} \frac{2^n}{n+5^n}$ comparison test.

$\leq \sum_{n=1}^{\infty} \frac{2^n}{5^n} = \sum_{n=1}^{\infty} \left(\frac{2}{5}\right)^n$ By geometric series, with $\left|\frac{2}{5}\right| < 1$.

$\Rightarrow$ Convergent!

(b) $\sum_{n=3}^{\infty} \frac{1}{\sqrt[3]{4n^3+4n+1}}$

$$\lim_{n\to\infty} \frac{\frac{1}{\sqrt[3]{4n^3+4n^2+1}}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n}{\sqrt[3]{4n^3+4n^2+1}} = \lim_{n\to\infty} \sqrt[3]{\frac{n^3}{4n^3+4n^2+1}}$$

$$= \lim_{n\to\infty} \sqrt[3]{\frac{1}{4+\frac{4}{n}+\frac{1}{n^3}}} = \sqrt[3]{\frac{1}{4}}$$

Then by limit comparison test. $\sum_{n=3}^{\infty} \frac{1}{\sqrt[3]{4n^3+4n^2+1}} \sim \sum_{n=3}^{\infty} \frac{1}{n}$,

which is divergent.

(c) $\sum_{n=1}^{\infty} (n^2+1)\sin\frac{1}{n^2}$ $\rightarrow$ positive series

Note that $\lim_{n\to\infty} (n^2+1)\sin\frac{1}{n^2}$

$$= \lim_{n\to\infty} n^2\sin\left(\frac{1}{n^2}\right) + \sin\left(\frac{1}{n^2}\right)$$

$$= \lim_{n\to\infty} n^2\sin\left(\frac{1}{n^2}\right) = \lim_{n\to\infty} \frac{\sin\left(\frac{1}{n^2}\right)}{\frac{1}{n^2}} \overset{u=\frac{1}{n^2}}{=} \lim_{u\to 0} \frac{\sin u}{u} = 1 \neq 0.$$

$\Rightarrow$ By test of divergence, divergent!

2. (Spring 2024, #7) Determine the convergence of each series. Circle your answer in each case and justify it carefully, clearly stating any tests you use.

(a) $\displaystyle\sum_{n=0}^{\infty}(\sin n)\left(\frac{n+3}{5n+7}\right)^{2n}$ Absolutely convergent Conditionally convergent Divergent

$$\sum_{n=0}^{\infty}|\sin(n)|\left(\frac{n+3}{5n+7}\right)^{2n}$$

$$\leq \sum_{n=0}^{\infty}\left(\frac{n+3}{5n+7}\right)^{2n} \quad (*)$$

Note that $\sqrt[n]{\left(\frac{n+3}{5n+7}\right)^{2n}} = \left(\frac{n+3}{5n+7}\right)^{2} \to \left(\frac{1}{5}\right)^{2} < 1$.

Then by root test, $(*)$ is convergent.

Thus, the original series is absolutely convergent.

(b) $\displaystyle\sum_{n=\cancel{0}\,2}^{\infty}\frac{1}{n(\ln n)^3}$ Convergent Divergent

integral test

Since $\frac{1}{x(\ln x)^3}$ continuous, positive and decreasing on $[2,\infty)$.

$$\int_2^{\infty}\frac{1}{x(\ln x)^3}\,dx$$

$$=\int_2^{\infty}\frac{1}{(\ln x)^3}\,d\ln x.$$

$$\overset{u=\ln x}{=} \int_{\ln 2}^{\infty}\frac{1}{u^3}\,du = \frac{1}{-2}\,\frac{1}{u^2}\Big|_{\ln 2}^{\infty}$$

$$= \frac{1}{2}\,\frac{1}{(\ln 2)^2} < \infty.$$

$\Rightarrow$ By integral test, the series is convergent.

3. (Fall 2023, #6) Determine whether the following series are conditionally convergent, absolutely convergent, or divergent. In each case, be sure to carefully justify your answer.

(a) $\displaystyle\sum_{n=1}^{\infty} \ln\left(\frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)}{2\cdot 5\cdot 8\cdot \ldots \cdot (3n-1)}\right)$

$$= -\sum_{n=1}^{\infty} \ln\left(\frac{2\times 5\times 8\times \cdots \times (3n-1)}{1\times 3\times 5\times \cdots \times (2n-1)}\right)$$

Note that $\displaystyle\sum_{n=1}^{\infty} \ln\left(\frac{2\times 5\times 8\times \cdots\times(3n-1)}{1\times 3\times 5\times \cdots (2n-1)}\right)$

$$\geq \sum_{n=1}^{\infty} \ln 2 = \infty.$$

$$\Rightarrow -\sum_{n=1}^{\infty} \ln\left(\frac{2\times 5\times 8\times \cdots \times(3n-1)}{1\times 3\times 5\times \cdots\times(2n-1)}\right) = -\infty.$$

$\Rightarrow$ Divergent.

(b) $\displaystyle\sum_{n=4}^{\infty} \frac{(-1)^n\cdot n^{3/2}}{n^3+n^2\cos(n)}$

$$|a_n| = \frac{n^{3/2}}{|n^3+n^2\cos n|} \leq \frac{n^{\frac{3}{2}}}{n^3-n^2} = \frac{n^{\frac{3}{2}}}{n^2(n-1)} = \frac{1}{n^{\frac{1}{2}}(n-1)} = \frac{1}{n^{\frac{3}{2}}}\cdot\frac{n}{n-1}$$

$$= \frac{1}{n^{\frac{3}{2}}}\left(1+\frac{1}{n-1}\right)$$

$$\leq \frac{2}{n^{\frac{3}{2}}}.$$

Note that by p-test, $\displaystyle\sum_{n=4}^{\infty}\frac{2}{n^{\frac{3}{2}}} < \infty$. since $\frac{3}{2} > 1$. Then, the series is absoluety convergent.

(c) $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n}{n\sqrt{\ln(n^2)}}$

Alternating test.

Since $\displaystyle\frac{1}{n\sqrt{\ln(n^2)}} \downarrow 0$.

We have $\displaystyle\sum_{n=2}^{\infty}\frac{(-1)^n}{n\sqrt{\ln(n^2)}}$ convergent.

However, $\displaystyle\sum_{n=2}^{\infty}\frac{1}{n\sqrt{\ln(n^2)}}$ is divergent.

By integral test $\displaystyle\int_2^{\infty}\frac{1}{x\sqrt{\ln x^2}}\,dx = \int_2^{\infty}\frac{1}{\sqrt{2\ln x}}\,d\ln x$

$$\overset{u=\ln x}{=} \int_{\ln 2}^{\infty}\frac{1}{\sqrt{2u}}\,du = \infty.$$

4. (Spring 2023, #7) Determine whether the following series are absolutely convergent, conditionally convergent, or divergent. Clearly state any test(s) you use and show that they are applicable. *Circle your answer for each part.*

(a) $\sum_{k=1}^{\infty} \frac{\sin k}{k^{1.26}}$ Absolutely convergent Conditionally convergent Divergent

$$\sum_{k=1}^{\infty} \left| \frac{\sin k}{k^{1.26}} \right| \leq \sum_{k=1}^{\infty} \frac{1}{k^{1.26}} < \infty, \quad \text{since } 1.26 > 1.$$

By p-test, absolutely convergent.

(b) $\sum_{n=1}^{\infty} \left(\frac{1}{n} - 1 \right)^n$ Absolutely convergent Conditionally convergent Divergent

$$\lim_{n\to\infty} \left(\frac{1}{n}-1\right)^n = \lim_{n\to\infty} (-1)^n \left(1-\frac{1}{n}\right)^n.$$

Since $\lim_{n\to\infty} \left(1-\frac{1}{n}\right)^n = e^{-1}$

We have

$$\lim_{k\to\infty} (-1)^{2k} \left(1-\frac{1}{2k}\right)^{2k} = e^{-1}$$

$$\lim_{k\to\infty} (-1)^{2k+1} \left(1-\frac{1}{2k+1}\right)^{2k+1} = -e^{-1}.$$

Thus, $\lim_{n\to\infty} (-1)^n \left(1-\frac{1}{n}\right)^n$ does not exist

$\Rightarrow$ Then, by the divergence test, the series is divergent.

5. (Fall 2022, #6) Determine whether the following series converge or diverge. Clearly state the method(s) you use.

(a) $\sum_{n=1}^{\infty} \frac{n^2+2}{\sqrt{n^7}+1}$ convergent.

limit comparison test. Let $a_n = \frac{n^2+2}{n^{\frac{7}{2}}+1}$

$b_n = \frac{n^2}{n^{\frac{7}{2}}} = \frac{1}{n^{\frac{3}{2}}}$.

$\lim_{n\to\infty} \frac{a_n}{b_n} = 1$. , By p-test $\sum_{n=1}^{\infty} b_n < \infty$. since $\frac{3}{2} > 1$.

Then $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{n^2+2}{\sqrt{n^7}+1} < \infty$.

(b) $\sum_{n=2}^{\infty} \frac{\ln n}{n^2}$ Convergent.

$f(x) = \frac{\ln x}{x^2}$ continuous, positive on $[2, \infty)$.

$f'(x) = \frac{\frac{1}{x}\cdot x^2 - 2x\ln x}{x^4} = \frac{1-2\ln x}{x^3} < 0$. for $x > 2$.

$\Rightarrow f(x)$ is decreasing on $[2, \infty)$.

Then, by integral test. since

$$\int_2^\infty \frac{\ln x}{x^2}\,dx = \int_2^\infty \ln x \, d\left(-\frac{1}{x}\right) = -\frac{\ln x}{x}\Big|_2^\infty - \int_2^\infty \frac{1}{x}\cdot\left(-\frac{1}{x}\right)dx.$$

$$= \frac{\ln 2}{2} + \int_2^\infty \frac{1}{x^2}\,dx = \frac{\ln 2}{2} + \left(-\frac{1}{x}\right)\Big|_2^\infty$$

$$= \frac{\ln 2}{2} + \frac{1}{2} < \infty.$$

$\Rightarrow$ The series is convergent.

Note that $\lim_{x\to\infty} \frac{\ln x}{x} \overset{\text{L'Hospital}}{=} \lim_{x\to\infty} \frac{\frac{1}{x}}{1} = 0$.

(c) $\sum_{n=1}^{\infty} \ln\left(1+\frac{1}{n}\right)$

Divergent.

Since $\ln(1+\frac{1}{n}) > 0$.

By limit comparison test.

$$\lim_{n\to\infty} \frac{\ln(1+\frac{1}{n})}{\frac{1}{n}} = \lim_{x\to 0} \frac{\ln(1+x)}{x}.$$

$$\overset{\text{L'Hospital}}{=} \lim_{x\to 0} \frac{\frac{1}{1+x}}{1} = 1.$$

$\Rightarrow$ Since $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$, by limit comparison test, $\sum_{n=1}^{\infty} \ln(1+\frac{1}{n})$ is also divergent.